How Much of Turkish Wordle Is Luck? A Multi-Baseline Skill/Luck Split Using Letter Frequency and Information Theory

Barış Cem Bayburtlu
Department of Software Engineering
Çankaya University
Ankara, Turkey
c2228009@student.cankaya.edu.tr

***Abstract*—We try to measure how much of winning at Turkish Wordle (*wordleturkce.bundle.app*) is luck and how much is skill. To do this we first pull the full word lists out of the live site: 2,936 unique answer words and 5,500 valid guesses. We then run four player models against every puzzle in the pool. A simple player that only follows the colored feedback (green, yellow, gray), with no strategy at all, already wins 96.97% of the games. The reason is that the feedback alone leaks a lot of information. But this player is also assumed to know every word in the dictionary, and that is not realistic for a human. To fix this we add two more models. The first is an entropy-greedy bot in the style of the *3Blue1Brown* lesson; it wins every puzzle and uses 3.59 guesses on average. The second is a human-like player who only knows the top K most common words, ranked by how often the bundle picks each word as the daily answer. With $K{=}1000$ (around 48% of daily puzzles), the split is about 34% luck and 66% skill. Inside that 66%, vocabulary alone explains about 95% and strategy adds the rest (about 5%). We also report the split for four other baselines. The headline number moves from 0.1% to 97% depending on which baseline we choose. All code, data, and figures are open source.**

***Index Terms*—Wordle, information theory, Shannon entropy, Turkish language, word games, skill versus luck, simulation, computational linguistics**

I. Introduction

Wordle was made by Josh Wardle in 2021 and bought by The New York Times in 2022. The player has six tries to guess a five-letter word. After each guess the game gives back one color per position: green (right letter, right place), yellow (right letter, wrong place), or gray (letter not in the word, with a small extra rule for repeated letters). The game got very popular and many local versions came out, including the Turkish-language *wordleturkce.bundle.app*, which uses its own list of Turkish five-letter words.

Players often ask: how much of winning is luck (the daily word happened to be easy) and how much is skill (the player picked smart guesses)? This is a fair question, but it is not so well defined. "Luck" only has a clear meaning once we say what "no skill" would look like. This paper answers the question with numbers for the Turkish version. We do five things:

1) Pull the full answer pool and guess dictionary out of the shipped JavaScript bundle (Section III);
2) Build four player models, from random to information-theoretic optimal (Sections IV, V);
3) Add a human-like player whose vocabulary is limited to the top K most common words (Section IV-D);
4) Compute the skill/luck split under five baselines and show how much the answer changes (Section VII); and
5) Look at per-word difficulty and the words that even the best bot finds hard (Section VIII).

The main result: under a realistic human baseline, Turkish Wordle wins are about 34% luck and 66% skill, and most of that skill (around 95%) is vocabulary, not strategy. This goes against the common view that picking the right opener is the most important thing.

II. Related Work

Wordle solvers got a lot of attention soon after the game came out. Sanderson's video [2] made the entropy-greedy approach widely known. It uses Shannon's information theory [1]: each turn the player picks the guess with the highest expected information gain, $H(g \mid C) = -\sum_p P(p \mid g, C)\log_2 P(p \mid g, C)$. Later work showed that the truly optimal English bot needs about 3.42 guesses on average and wins every game by doing full game-tree search [3]. A recent peer-reviewed paper by Aladaileh et al. [8] confirms again that entropy beats simple letter-distribution heuristics on English Wordle.

For Turkish Wordle, Akbulut and Senturk [13] already did the first careful comparison of LSTM, MaxEnt, and rule-based solvers on 1,000 Turkish target words. They report that linguistic features—dense CVCVC templates and low lexical frequencies in particular—predict how hard a word is. Our paper goes in a different direction. We do not compare solvers; we look at the luck/skill split. We also use the full 2,936-word solution pool instead of a sample of 1,000, and we add a human player with a limited vocabulary.

Source code, data, and figures are available at https://github.com/byigitt/what-percentage-wordle-is.

The vocabulary-limited idea has a close relative in the HSW model of Qiu and Zhong [14], who use a vocabulary-population (VP) component based on a normal-distribution assumption to simulate heterogeneous English players. Our model is different: we do not assume any distribution. Instead we rank words by how often the deployed bundle picks them as the daily answer.

For games in general, the skill-vs-luck question has been studied more formally. The most rigorous work is on poker [5], which uses a predictive-likelihood argument. We follow a simpler counterfactual idea from Mauboussin [4]: a game outcome counts as skill if the skilled agent wins where the baseline agent would lose, and as luck if both win or both lose. For English Wordle, Dilger [15] applies information theory to a large sample of player data and uses the share of one-guess solves to estimate a cheating rate. Our work is different: we report the skill/luck split under several human baselines, not a single score against one bot.

We are not aware of earlier work that gives a multi-baseline skill/luck split for Wordle in any language together with a vocabulary-limited human model built from real data. Turkish has its own structure: it is agglutinative and prefers CVCVC syllables, so the letter and position counts are not the same as English. This matters for opener choice and for which words turn out to be hard.

III. Data and Methodology

A. Word Lists

The game runs fully in the browser. The whole word list ships inside one JavaScript file at https://wordleturkce.bundle.app/static/main.2023-04-23.js. We find two arrays in it:

- `La`: 11,470 entries, but only 2,936 unique. The daily answer is $La[i \bmod 11470]$, where i is the day index. The same word can appear many times—for example `buruk` and `selam` each appear 8 times, while rare words appear once. We read this repetition as a commonality signal: the curator made common words more likely to be the daily answer.
- `Ta`: 5,531 entries, 5,499 unique. These count as valid guesses but never appear as the answer.

After we drop duplicates, the full guess dictionary has 5,500 unique words. The daily answer comes from the 2,936 unique solutions, with the La multiplicities used as weights.

B. Feedback Function

We implement the standard Wordle feedback rule, including the case with repeated letters. It is a deterministic function $\text{feedback} : \Sigma^5 \times \Sigma^5 \to \{0, 1, 2\}^5$, where 0, 1, and 2 stand for gray, yellow, and green. We assign greens first. Then we walk through the non-green positions of the guess from left to right and mark a yellow when the answer still has an unmatched copy of that letter. We encode each five-cell pattern as a base-3 integer in $[0, 242]$ so that we can index it directly.

C. Letter Frequency Analysis

Over the weighted 11,470-entry solution distribution, the five most common letters are A (13.04%), E (8.52%), İ (6.90%), K (6.81%), and R (6.09%). Together they cover 41% of all letter occurrences. By position:

TABLE I
MOST FREQUENT LETTERS BY POSITION (TOP 5 EACH).

Position	Top 5 letters (with counts)
1	K (1336), S (1121), T (841), B (751), M (714)
2	A (3199), E (1881), İ (1142), O (938), U (782)
3	R (1435), L (1183), K (799), N (757), M (747)
4	A (2308), İ (1587), E (1530), I (868), U (843)
5	K (1367), N (1189), R (1070), A (1052), E (971)

This pattern matches Turkish phonotactics. Turkish words use a canonical $(C)V(C)$ syllable template [12], and CVCVC is one of the most common five-letter shapes. Vowels are mostly in positions 2 and 4 and consonants in positions 1, 3, and 5.

IV. Player Models

We use four player models. All of them follow the standard rules: up to six guesses, and each guess must be a valid five-letter word from the dictionary.

A. Pure Random ($\mathcal{P}_{\text{rand}}$)

At each turn this player picks a random word from the 5,500-word dictionary and ignores all feedback. This is the zero-information baseline. The expected win rate in six guesses is $1 - (1 - 1/5500)^6 \approx 0.109\%$.

B. Consistent Random ($\mathcal{P}_{\text{cons}}$)

This player keeps the set of candidates that are still consistent with the feedback so far. At each turn it picks a random word from that set. It follows the rules perfectly but uses no strategy beyond that.

C. Frequency Bot ($\mathcal{P}_{\text{freq}}$)

At each turn this bot recomputes positional letter frequencies on the current candidate set. Each candidate w gets a score $\text{score}(w) = \sum_i \mathbb{1}[w[i] \notin w[0{:}i]] \cdot \text{posfreq}[i, w[i]]$, which sums the contributions of the *distinct* letters only. The bot picks the candidate with the highest score.

D. Vocabulary-Limited Human ($\mathcal{P}_{\text{hum},K}$)

This model represents a casual human player by a vocabulary size K. We rank words by how often `La` repeats them (the commonality proxy), and break ties alphabetically. The top K are the player's vocabulary. At each turn, the player picks a random word from the intersection of the remaining candidates and the vocabulary. If that intersection is empty, the player picks a random word from the vocabulary as a forced "probe" guess.

TABLE II
PERFORMANCE OF ALL FOUR PLAYER MODELS ON THE 2,936-WORD SOLUTION POOL (UNIFORM SAMPLING).

Player	Win rate	Avg. guesses	Opener
$\mathcal{P}_{\mathrm{rand}}$ (pure random)	0.10%	4.33	—
$\mathcal{P}_{\mathrm{hum},1000}$ (K=1000)	33.92%	3.84	—
$\mathcal{P}_{\mathrm{cons}}$ (consistent)	96.97%	4.21	—
$\mathcal{P}_{\mathrm{freq}}$ (frequency)	99.18%	3.77	`salik`
$\mathcal{P}_{\mathrm{ent}}$ (entropy)	**100.00%**	**3.59**	`merak`

V. ENTROPY-GREEDY BOT

To get a tight skill ceiling we add an entropy-greedy bot $\mathcal{P}_{\mathrm{ent}}$ in the style of [2]. For a guess g and candidate set C, the partition that g creates on C is $\{C_p : p \in \{0, \dots, 242\}\}$ where $C_p = \{a \in C : \mathrm{feedback}(g, a) = p\}$. The expected information gain is the entropy of this partition:

$$H(g \mid C) = -\sum_p \frac{|C_p|}{|C|} \log_2 \frac{|C_p|}{|C|}. \tag{1}$$

The bot picks $g^\star = \arg\max_g H(g \mid C)$ and, when several guesses are tied, prefers a guess that is itself in C.

A. Implementation

A direct implementation needs $O(|G| \cdot |C|)$ feedback evaluations per turn, which is too slow when we simulate thousands of puzzles. To speed this up we precompute a pattern matrix $M \in \{0, \dots, 242\}^{|G| \times |A|}$ once. We store the encoded feedback for every guess–answer pair as a `uint8`. The matrix takes about 8 MB and is built in about 0.5 s using vectorized NumPy [9] operations that work in parallel over letters.

To check the fast feedback function we compared it with a plain Python reference on 5,000 random word pairs. We found 0 mismatches.

B. Results

The best entropy opener is `merak` with $H = 5.81$ bits. The ten best openers by entropy are `merak, kenar, kamer, kelam, kalem, malik, karne, katre, keman, karni`. Interesting note: mid-frequency liquid consonants (R, L, N, M) win over the much more frequent letter A, which dominated the frequency-score top picks.

On all 2,936 puzzles, the entropy bot wins 100.00% of the time, with 3.59 guesses on average. It never reaches turn six. The turn histogram is $\{1 : 1, 2 : 84, 3 : 1221, 4 : 1455, 5 : 175\}$.

VI. SIMULATION RESULTS

Table II shows the main numbers. Fig. 1 shows the turn distribution for each model. The most surprising thing is the gap between $\mathcal{P}_{\mathrm{rand}}$ (0.1%) and $\mathcal{P}_{\mathrm{cons}}$ (96.97%). If you just follow the feedback properly, you already get almost the whole win rate. Strategy on top of that only adds a small amount.

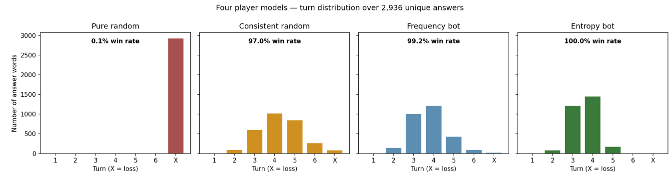

Fig. 1. Turn counts for the four player models on 2,936 puzzles. Pure random almost always runs out of guesses. Consistent random and the frequency bot have long tails. The entropy bot is tightly grouped and never goes past five guesses.

TABLE III
WIN RATES OF THE VOCABULARY-LIMITED HUMAN MODEL AND THE SHARE OF DAILY PUZZLES ITS VOCABULARY COVERS, AS A FUNCTION OF K.

K	Uniform win %	Weighted win %	Coverage %
200	6.81	10.92	10.92
500	17.00	25.74	25.79
1000	33.92	47.39	47.59
1500	50.72	67.00	67.50
2000	67.13	83.68	84.93
2500	83.58	94.46	96.20
2936	97.21	97.20	100.00

A. Vocabulary Sensitivity

Table III and Fig. 2 show how the win rate depends on the vocabulary size K. The empirical fit

$$P(\mathrm{win} \mid K) \approx \mathrm{coverage}(K) \cdot 0.97 \tag{2}$$

holds for every K we tested, within sampling noise. Equation 2 says a simple thing: *knowing the word is almost necessary and sufficient to win*. If the day's word is in the player's vocabulary, consistent-random play wins with probability about 0.97. Otherwise the player loses.

VII. SKILL/LUCK DECOMPOSITION

We use a counterfactual decomposition. The entropy bot's wins count as *luck* when the baseline player would have won too, and as *skill* when the baseline player would have lost. For a baseline $\mathcal{P}_b$ with win rate w_b and a ceiling $\mathcal{P}_{\mathrm{ent}}$ with win rate $w_c = 1$, the luck share is w_b/w_c and the skill share is $(w_c - w_b)/w_c$.

Table IV and Fig. 3 show the split. The headline answer depends entirely on the baseline we pick. We argue that the most realistic baseline for an average human is $\mathcal{P}_{\mathrm{hum},K\approx1000}$, since informal estimates put active Turkish five-letter vocabulary around this size for casual players. Under this baseline, $\approx 66\%$ **of Turkish Wordle wins come from skill and** $\approx 34\%$ **come from luck.**

A. Skill is Mostly Vocabulary

Equation 2 also tells us that the skill share itself splits into two parts. Going from $\mathcal{P}_{\mathrm{hum},1000}$ to $\mathcal{P}_{\mathrm{cons}}$ (perfect vocabulary, no strategy) adds 63 points of win rate (33.92% to 96.97%). Going from $\mathcal{P}_{\mathrm{cons}}$ to $\mathcal{P}_{\mathrm{ent}}$ (strategy on top of perfect vocabulary) adds only 3 points more (96.97% to 100.00%). So out

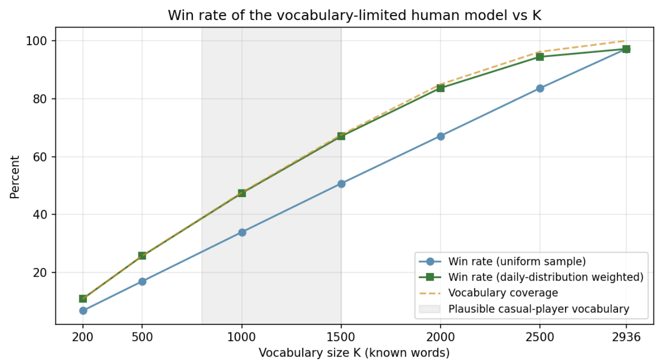

Fig. 2. Win rate of the vocabulary-limited human model as a function of vocabulary size K. The weighted curve is almost identical to the vocabulary coverage curve, which matches Eq. 2. The shaded band is the plausible range for typical casual players.

TABLE IV
SKILL/LUCK SPLIT UNDER FIVE DIFFERENT BASELINES. THE RESULT CHANGES BY ALMOST TWO ORDERS OF MAGNITUDE.

Baseline $\mathcal{P}_b$	w_b	Luck %	Skill %
Pure random	0.10%	0.1%	99.9%
Human (K=500)	17.00%	17.0%	83.0%
Human (K=1000)	**33.92%**	**33.9%**	**66.1%**
Human (K=2000)	67.13%	67.1%	32.9%
Consistent random	96.97%	97.0%	3.0%

of the 66-point skill gap, vocabulary explains about 95% and strategic word selection explains about 5%.

This goes against the usual story in popular Wordle posts, which focuses on opener choice and information-theoretic optimization. Our result suggests that *lexical knowledge* matters much more than strategic sophistication for who wins more often.

VIII. Per-Word Difficulty

Fig. 4 shows how many turns the entropy bot needs to solve each of the 2,936 puzzles. The most common turn count is 4. The bot never reaches turn 6. The puzzles that take 5 turns (175 words) are the structurally hardest ones.

When we look inside the 5-turn group we see a common pattern: large families of look-alike words that differ in only one position, for example `KAPAK / KAÇAK / KAVAK / KAZAK`. Once the candidate set shrinks to such a family, even an optimal bot has to test the remaining words one by one because each guess tells it about one position only. This is an irreducible source of variance. No strategy can remove it, and an ideal human would face the same problem.

This kind of luck is *structural*, and is different from the player-level luck we measured earlier. Even a perfect-information player has to spend more turns when the day's word lives in a dense look-alike family.

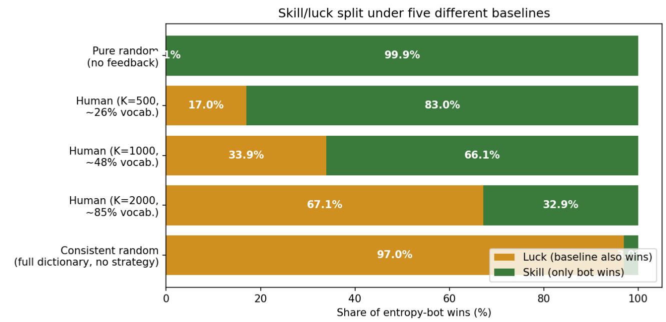

Fig. 3. Skill and luck shares of entropy-bot wins under five baselines. The answer to "how much of Wordle is luck" moves from 0.1% to 97% just based on which baseline we pick.

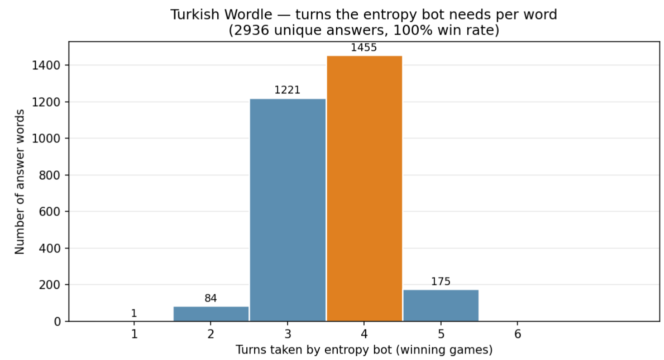

Fig. 4. Turn counts of the entropy-greedy bot on all 2,936 solutions. No puzzle needs 6 turns; the hardest 175 take 5.

IX. Discussion

A. Definitional Sensitivity

The main methodological point of this work is that the popular "$X\%$ luck, $Y\%$ skill" framing is not well defined without a baseline. The same data, looked at under five reasonable baselines, gives skill shares anywhere between 3% and 99.9%. We therefore think single-number claims of this kind should be avoided in both popular and academic discussion of game design.

B. Vocabulary as the Dominant Skill

The near-identity between vocabulary coverage and win rate (Eq. 2) points to a finding that we think has not been discussed enough: Wordle is mostly a vocabulary test. Strategy helps, but only a little. A player who wants a higher win rate is better off learning more words than switching to a fancier opening word. This matches independent observational work by Liang et al. [11], who show that real Wordle players narrow candidates using semantic, orthographic, and morphological similarities, not information-theoretic optimality.

C. Information-Theoretic Bound

The Shannon entropy of the weighted daily distribution is 11.36 bits. The theoretical ceiling per guess is $\log_2 243 = 7.92$ bits. The entropy bot pulls out about 3.16 bits per guess on average (the same as needing 3.59 guesses to remove 11.36 bits of uncertainty). That is around 40% of the per-guess ceiling. The gap has two sources: (i) the greedy choice is not the same as full lookahead, and (ii) every guess must be a real word, not just any five-character string.

D. Limitations

The vocabulary model uses bundle multiplicity as a proxy for commonality. A more rigorous human model would calibrate against an external Turkish corpus and would also include imperfect feedback tracking, especially for words with repeated letters. Stronger solver models, such as the Bertsimas–Paskov-style full-tree search [3], would push the skill ceiling a tiny bit higher but would not change the overall split.

X. Conclusion

We have presented a multi-baseline analysis of skill versus luck in Turkish Wordle, built on a full extraction of the deployed game's solution and dictionary lists. Four player models were simulated on all 2,936 puzzles: the entropy-greedy ceiling wins every game, while a realistic vocabulary-limited human (K=1000) wins about 34%. Under this baseline, the skill share of wins is about 66%, and about 95% of that is vocabulary while only about 5% is strategy. We stress that the headline split is very sensitive to the baseline and we report a five-baseline table to back up this point.

All code, simulation outputs, intermediate data, and reproduction steps are available as open source at the project repository.

Reproducibility

The source code, the extracted word lists, and the full simulation outputs are released under a permissive license at the project's open-source repository. The whole pipeline runs end-to-end in under three minutes on a normal laptop, using only NumPy [9] and Matplotlib [10].

References

[1] C. E. Shannon, "A mathematical theory of communication," *Bell System Technical Journal*, vol. 27, no. 3, pp. 379–423, 1948.

[2] G. Sanderson (3Blue1Brown), "Solving Wordle using information theory," YouTube video, 2022. https://www.youtube.com/watch?v=v68zYyaEmEA

[3] D. Bertsimas and A. Paskov, "An Exact Solution to Wordle," *Operations Research*, vol. 73, no. 3, pp. 1384–1394, 2025.

[4] M. J. Mauboussin, *The Success Equation: Untangling Skill and Luck in Business, Sports, and Investing*. Harvard Business Review Press, 2012.

[5] S. D. Levitt and T. J. Miles, "The Role of Skill Versus Luck in Poker: Evidence from the World Series of Poker," *Journal of Sports Economics*, vol. 15, no. 1, pp. 31–44, 2014.

[6] J. Wardle, "Wordle," originally published 2021; https://www.nytimes.com/games/wordle/index.html.

[7] "Wordle Türkçe," https://wordleturkce.bundle.app/, accessed 2026.

[8] T. Aladaileh, D. Stephens, M. Alqaisi, and C. Wu, "Solving Wordle Using Information Theory," *Northeast Journal of Complex Systems*, vol. 8, no. 1, art. 6, 2026. https://orb.binghamton.edu/nejcs/vol8/iss1/6/

[9] C. R. Harris *et al.*, "Array programming with NumPy," *Nature*, vol. 585, no. 7825, pp. 357–362, 2020.

[10] J. D. Hunter, "Matplotlib: A 2D graphics environment," *Computing in Science & Engineering*, vol. 9, no. 3, pp. 90–95, 2007.

[11] G. Liang, A. Kabbara, C. Liu, R. Luo, K. Kim, and M. Guerzhoy, "Semantic, Orthographic, and Phonological Biases in Humans' Wordle Gameplay," arXiv preprint arXiv:2411.18634, 2024. https://arxiv.org/abs/2411.18634

[12] A. Göksel and C. Kerslake, *Turkish: A Comprehensive Grammar*. London: Routledge, 2005.

[13] S. Akbulut and İ. F. Senturk, "An analytical study of artificial intelligence methods in the Turkish Wordle game: LSTM, entropy, and rule-based approaches," *Entertainment Computing*, vol. 57, art. 101117, May 2026, doi: 10.1016/j.entcom.2026.101117.

[14] T. Qiu and Y. Zhong, "Simulation and analysis of Wordle based on the HSW model," in *Frontiers in Artificial Intelligence and Applications*, vol. 376, IOS Press, 2024, pp. 167–174.

[15] J. P. Dilger, "Wordle: A microcosm of life. Luck, skill, cheating, loyalty, and influence!" arXiv preprint arXiv:2309.02110, 2023. https://arxiv.org/abs/2309.02110